It will take up to five minutes for the stack to be created. Log in to the AWS Management Console, and navigate to the AWS CloudFormation service to monitor the process of the CloudFormation stack named `url2png`.

We're using the length of the SQS queue to scale the number of EC2 instances. Because the number of messages in the queue does not correlate with the number of EC2 instances processing messages from the queue, it is not possible to use a target-tracking policy. Therefore, you will use a step-scaling policy in this scenario, as illustrated here.

Listing 17.7 Monitoring the length of the SQS queue

```
# [...]
HighQueueAlarm:
  Type: 'AWS::CloudWatch::Alarm'
  Properties:
    EvaluationPeriods: 1                                ①
    Statistic: Sum                                      ②
    Threshold: 5                                        ③
    AlarmDescription: 'Alarm if queue length is higher than 5.'
    Period: 300                                         ④
    AlarmActions:                                       ⑤
    - !Ref ScalingUpPolicy
    Namespace: 'AWS/SQS'                                ⑥
    Dimensions:                                         ⑦
    - Name: QueueName
      Value: !Sub '${SQSQueue.QueueName}'
    ComparisonOperator: GreaterThanThreshold            ⑧
    MetricName: ApproximateNumberOfMessagesVisible      ⑨
# [...]
```

① Number of time periods to evaluate when checking for an alarm

② Sums up all values in a period

③ Alarms if the threshold of five is reached

④ Uses a period of 300 seconds because SQS metrics are published every five minutes

⑤ Increases the number of desired instances by one through the scaling policy

⑥ The metric is published by the SQS service.

⑦ The queue, referenced by name, is used as the dimension of the metric.

⑧ Alarms if the sum of the values within the period is greater than the threshold of five

⑨ The metric contains an approximate number of messages pending in the queue.

The CloudWatch alarm triggers a scaling policy. The scaling policy shown in the following listing defines how to scale. To keep things simple, we are using a step-scaling policy with only a single step. Add additional steps if you want to react to a threshold breach in a more fine-grained way.